



Spaceborne Lidar

Quick Look Products

GEDI Data Products:

The Global Ecosystem Dynamics Investigation

(GEDI) mission is a space-based laser altimeter designed to measure vertical structure and topography of Earth's forests with high resolution. GEDI plays a crucial role in advancing our ability to understand atmospheric and water processes, biodiversity, and habitat. Billions of full-waveform observations are collected each year providing 3-dimensional information to forest, weather, water, and fire risk management.

How do I access the data?

GEDI L1-2 are maintained by LP DAAC, and all other data by the ORNL DAAC. All products and documentation are stored in the [EarthData catalog](#).

Where can I find more information?

The GEDI Mission webpage, [product specific](#) user guides, associated and SDS Data Dictionaries, and Algorithm Theoretical Based Documents (ATBDs) have the most updated information on the latest product version. New product levels and derived datasets are forthcoming.



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Data Product	GEDI01_A-TX Transmitted Waveform Fitted Parameters	GEDI01_B Footprint Level Geolocated Waveform Data	GEDI02_A Footprint Level Elevation and Height Metrics	GEDI02_B Footprint Level Canopy Cover and Vertical Profile Metrics
	GEDI01_A-RX Received Waveform Fitted Parameters			
Datasets	Raw waveform	Geolocated waveform	Ground elevation, canopy top height, relative height (RH) metrics	Canopy Cover Fraction (CCF), CCF profile, Plant Area Index (PAI), PAI profile, Plant Area Volume Density (PAVD) profile, Foliage Height Diversity (FHD)
Resolution	25 m (~82 ft) diameter			
Spatial Coverage	N: 54, S: -54, E: 180, W: -180			
*Temporal Extent	2019-04-04 to 2023-03-16 2024-04-26 to Present			
Temporal Resolution	Varies			
Algorithm Theoretical Based Document	L1A-2A : Transmit and Receive Waveform Interpretation and Generation of L1A and L2A Products	L1B : Waveform Geolocation for L1 and L2 Products	L1A-2A : Transmit and Receive Waveform Interpretation and Generation of L1A and L2A Products	L2B : Footprint Canopy Cover and Vertical Profile Metrics
Archive / Data Dictionary	Not publicly available	NASA LP DAAC		
Data Format	HDF5			

* temporal extents/aggregations may update with new data versions.

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	GEDI03	GEDI04_A	GEDI04_B	GEDI04_C
Data Product	<u>Gridded Level 2 Land Surface Metrics</u>	<u>Footprint Level Above Ground Biomass Density (AGBD)</u>	<u>Gridded Above Ground Biomass Density (AGBD)</u>	<u>Footprint Level Waveform Structural Complexity Index (WSCl)</u>
Datasets	Mean and standard deviation of RH100 and ground elevation, counts of laser footprints within each grid	AGBD, AGBD standard error and prediction intervals, model prediction and standard error	Mean, variance, standard error of AGBD, sample and cluster counts, plant functional type prediction stratum used, and model type	WSCl, WSCl profile, WSCl horizontal with prediction intervals for each
Resolution	1 km (~0.6 mi) grids	25 m (~82 ft) diameter	1 km (~0.6 mi) grids	25 m (~82 ft) diameter
Spatial Coverage	N: 52, S: -52, E: 180, W: -180	N: 55.8, S: -53, E: 180, W: -180	N: 52, S: -52, E: 180, W: -180	N: 55.8, S: -53, E: 180, W: -180
Temporal Extent	2019-04-18 to 2023-03-22	2019-04-17 to 2025-03-19*	2019-04-18 to 2023-03-16*	2019-04-17 to 2025-03-19*
Temporal Resolution	Varies			
Algorithm Theoretical Based Document	L3: Gridded Land Surface Metrics	L4A: Footprint Above Ground Biomass Density	L4B: Gridded Biomass Product	L4C: Footprint Waveform Structural Complexity Index
Archive / Data Dictionary	NASA ORNL DAAC			
Data Format	GeoTIFF	HDF5	GeoTIFF	HDF5

** temporal extents/aggregations may update with new data versions.*

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Data Product (Derived)	<u>Level 4B Country-level Summaries of Above Ground Biomass</u>	<u>Pantropical Forest Height and Biomass from GEDI and TanDEM-X Data Fusion</u>	<u>Global Vegetation Height Metrics from GEDI and ICESat-</u>	<u>Gridded GEDI Vegetation Structure Metrics and Biomass Density at Multiple Resolutions</u>
Datasets	Total mean and standard error, and standard error percent AGBD in Mg ha ⁻¹ and Pg C units	Mean biomass and uncertainty, mean canopy height and uncertainty by model and forest disturbance	RH50, RH75, RH90, RH98, sample count	AGBD, CCF, ground elevation, PAI, PAI profile, PAVD profile, FHD, RH50, RH95, RH98, PAI and PAVD fraction (all with options across multiple processes applied)
Resolution	Country level	25 m, 100 m	100 m, 200 m, 500 m, 1000 m	1 km, 6 km, and 12 km grids
Spatial Coverage	N: 52, S: -52, E: 180, W: -180	Mexico, Gabon, French Guiana, Amazonia	N: 90, S: -90, E: 180, W: -180	N: 52.2, S: -52.2, E: 180, W: -180
Temporal Extent	2019-04-18 to 2023-03-16	2011-01-06 to 2021-08-18*	2019-01-01 to 2022-12-31*	2019-04-17 to 2023-03-16*
Temporal Resolution	Varies			
Algorithm Theoretical Based Document	Country estimates of mean above ground biomass density and total above ground biomass stocks	<u>Mapping large-scale pantropical forest canopy height by integrating GEDI lidar and TanDEM-X InSAR data</u>		
Archive / Data Dictionary	NASA ORNL DAAC			
Data Format	GeoTIFF			

* temporal extents/aggregations may update with new data versions.